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GAs STUDENT SUBMISSION COVER PAGE

	SURNAME & INITIALS	STUDENT NO.	SIGN	DATE
	Zungu N	202375568	N ZUNGU	2026/05/13

Topic: Smart Airport Luggage Sorting and Tracking System Description

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Appendix A

1. AI Tool Use Disclosure

For each AI tool that you have used, provide a brief, high-level summary:

1.	Tool Name and Version:	DeepSeek
	Purpose:	For generating a figure of the layout plan
	Scope of Use	

2.	Tool Name and Version:	ChatGBT
	Purpose:	For finding the right papers based on my topic
	Scope of Use	

**Add as many tables as you require or delete tables as appropriate.*

2. AI Tool Use Detailed Disclosure by Category

For every instance where you used an AI tool, you must provide a detailed explanation.

A. AI as a Writing Tool (e.g., drafting, summarising, paraphrasing, editing, proofreading, image generation)

<i>Requirement</i>	<i>Explanation</i>
<i>Specific use</i>	Mathematical and Analytical Considerations
<i>Input/Prompt</i>	<i>prompt</i>
<i>Validation process</i>	
<i>Required evidence</i>	

**Add as many tables as you require.*

B. AI as an Idea Generation Tool (e.g., brainstorming, outlining, topic selection, structural design)

<i>Requirement</i>	<i>Explanation</i>
<i>Specific use</i>	Summarizing Engineering Knowledge, and the improve English I used
<i>Input/Prompt</i>	<i>input</i>
<i>Validation process</i>	
<i>Required evidence</i>	

Abstract

This report carries on from Assignment 1, where the carousel sorter with RFID tracking and SMS passenger notification was picked as the best design for a smart airport luggage sorting and tracking system [1]. The carousel sorter was chosen over the cross-belt option cause its simpler, cheaper, easier to maintain, and more suitable for developing environments. The system improves baggage movement from check-in to aircraft loading and from aircraft arrival to baggage reclaim. This detailed design covers system configuration, main parts, integration with existing airport infrastructure, safety stuff, and a basic testing plan [1].

The design also addresses practical challenges common in developing environments, such as unstable power supplies, limited budgets, low local maintenance skills, and weak internet connectivity. The system is built to keep working even when the internet connection is bad.

This design follows the detailed design approach I was taught in the 5EEE421 lectures. The final system had to be clear, safe, reliable, maintainable, and ready to implement [1],[2].

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1. Introduction

Handling baggage at airports is a big deal cause passengers expect their luggage to arrive safely and on time. In Assignment 1, two sorting options were compared: a carousel sorter and a cross-belt sorter. The carousel sorter with RFID and SMS got picked because it strikes a good balance between cost, performance, simplicity, and maintenance [1],[8]. The previous design spec needed the system to handle at least 1500 bags per hour, track bags in under 5 seconds, notify passengers in under 60 seconds, and fix problems in under 30 minutes [1].

The point of this Part 2 report is to give the detailed design of the chosen system. That means explaining how the system will be set up, what parts will be used, how it will hook into existing airport infrastructure, and how safety will be handled [8]. Detailed design matters because it turns the chosen concept into something that can be installed, tested, used, and maintained [3]. Detailed design must consider function, safety, standards, picking components, drawings, documentation, cost, and testing [3]

2. Selected Design Solution

The chosen design is a carousel baggage sorter with RFID tracking and SMS notification. In this system, when a passenger checks in, their bag gets an RFID tag. Then the bag moves through conveyors, screening, sorting, loading, and arrival areas. RFID readers scan the bag at fixed checkpoints, while the PLC controls motors, sensors, pushers, and safety devices. When the baggage status changes, the system updates the database and sends an SMS to the passenger.

The carousel sorter works well because it uses one main conveyor drive, has fewer moving parts than a cross-belt sorter, and uses simpler rule-based control. In Assignment 1, the carousel sorter was described as running at about 1 m/s with a capacity of around 2700 bags per hour, which is above the needed 1500 bags per hour [1].

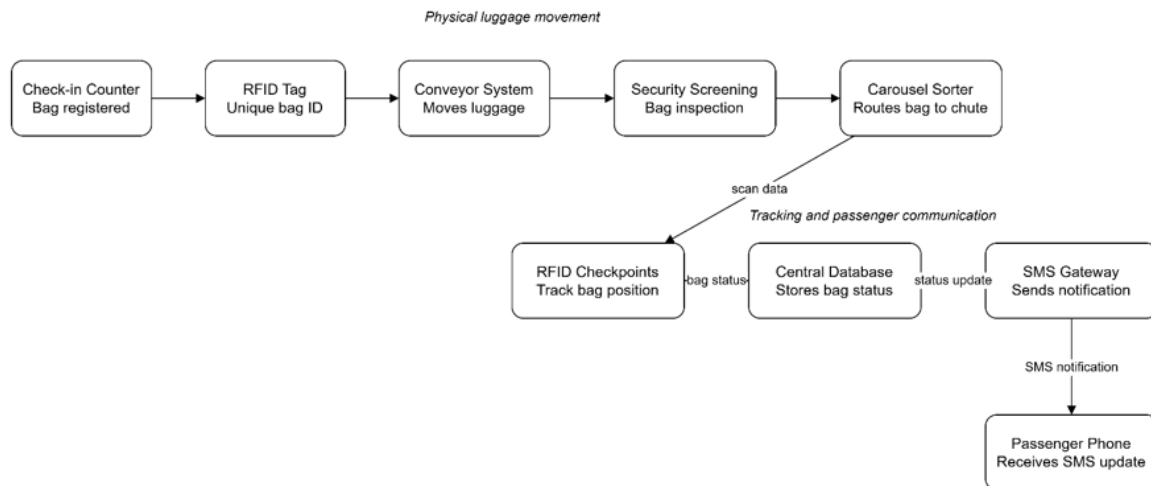


Figure 1 Proposed system block diagram

3. System Configuration and Main Components

The system is split into five main subsystems: mechanical handling, sensing and tracking, control, communication, and user interface.

3.1 Mechanical Handling Subsystem

The mechanical subsystem moves and sorts of baggage. It includes the main conveyor belt, carousel sorter, chutes, pusher mechanism, guide rails, and support frame. The carousel sorter sends bags to the right chute using a pusher mechanism controlled by the PLC.

main components:

Table 1 Main components

Component	Proposed specification	Purpose
Conveyor belt	1 m/s nominal speed	Moves bags through the system
Main motor	3-phase AC induction motor, 1.5–3 kW	Drives the conveyor
Variable Speed Drive	380–415 V AC, motor-rated	Controls conveyor speed
Carousel/pusher unit	Pneumatic or motor-driven pusher	Pushes the bag into the correct chute
Chutes	Steel frame with rollers	Directs bags to the loading/reclaim area
Frame	Mild steel, powder-coated	Supports system

The design should use a variable speed drive because it lets you adjust the conveyor speed based on how much baggage there is [9]. This helps save energy when there's not a lot of bags coming through.

3.2 RFID Tracking Subsystem

The RFID part handles identifying each bag on its own. RFID is good cause unlike barcodes, you don't need a direct line of sight, and it scans fast when the bag passes a checkpoint [6]. IATA is big on tracking baggage the whole way through, and RFID is one of the main ways airports use to keep tabs on luggage and stop bags from getting lost [6].

Table 2 Tracking Subsystem Components

Component	Proposed specification	Purpose
RFID tag	Passive UHF RFID baggage tag	Unique bag identification
RFID reader	UHF portal reader	Reads tag at checkpoint
Antenna	Directional antenna	Improves scanning area
Checkpoints	7 scan zones	Tracks baggage movement
Database link	Ethernet/LAN	Sends scan data to server

The seven RFID zones should go at check-in exit, screening entry, screening exit, sorter entry, sorter exit, loading bay, and arrival carousel. This layout follows the tracking plan from Assignment 1, where RFID readers were put at key baggage handling points [1].

3.3 Control Subsystem

The control subsystem runs on a PLC cause airport baggage systems need reliable and deterministic control. The PLC gets inputs from sensors and RFID events, then controls the conveyor motor, pusher, alarms, and emergency stops.

Table 3 Control Subsystem Components

Component	Proposed specification	Purpose
PLC	Siemens S7-1200 or equivalent	Main control
Digital inputs	24 V DC	Sensors and switches
Digital outputs	24 V DC relay/transistor	Motor contactors and alarms
HMI screen	7-inch touch panel	Operator display
Power supply	24 V DC industrial supply	PLC and sensors
Control panel	IP-rated enclosure	Electrical protection

3.4 Communication and Database Subsystem

The communication subsystem connects the RFID readers, PLC, database server, staff dashboard, and SMS gateway. The system should use a local area network so that the baggage system can still operate even if the internet is unstable [7]. This supports the Assignment 1 idea that local processing is better for developing environments where the internet and power can be unreliable [1].

Table 4 Database Components

Component	Proposed specification	Purpose
Local server	SQL database server	Stores baggage status
LAN switch	Industrial Ethernet switch	Connects devices
SMS gateway	Bulk SMS API or GSM modem	Sends passenger updates
Dashboard PC	Existing airport workstation	Staff monitoring
Backup power	UPS for the server and network	Keeps tracking alive during a short outage

3.5 Passenger and Staff Interface

The passenger interface uses SMS because it works on smartphones and basic phones alike. Passengers don't get to install any app either. The staff interface is a dashboard that shows baggage status, RFID scan points, missed scans, faults, and emergency alarms.

example SMS messages:

"Your bag has been checked in successfully."

"Your bag has passed security screening."

"Your bag is ready for collection at Carousel 3."

4. Integration with Existing Airport Infrastructure

The system should be installed in phases, so the airport does not need to replace everything at once. This is important because Assignment 1 identified cost, unstable funding, and existing barcode systems as major issues in developing environments [1] [8].

4.1 Physical Integration

The carousel sorter will be installed after the security screening area and before the baggage make-up/loading area. Existing conveyors can be reused where possible. RFID portal frames will be mounted above or beside the conveyor at fixed points [8]. The system must be installed without blocking emergency walkways or maintenance.

Figure 2: Layout Plan

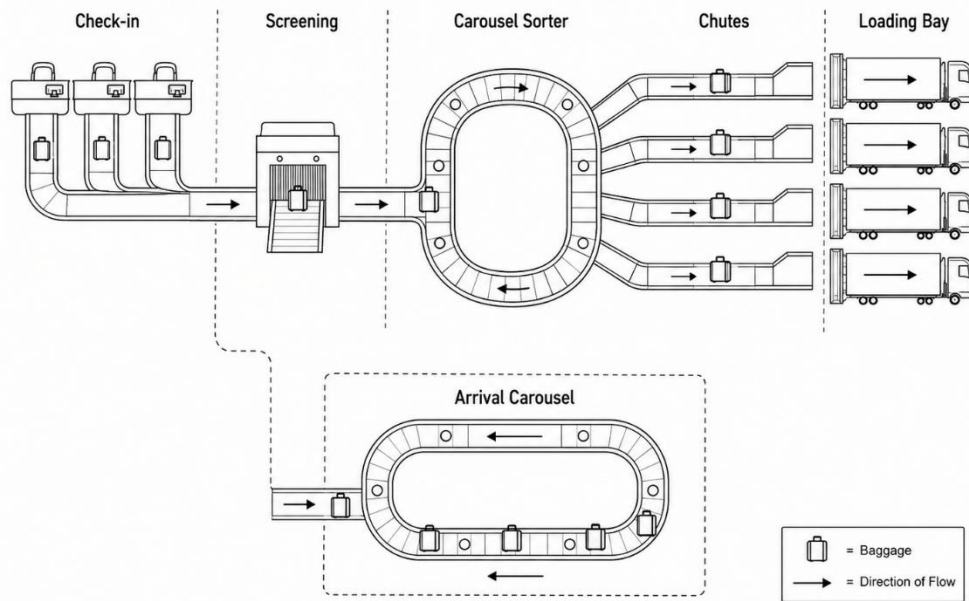


Figure 2 Layout plan

4.2 Electrical Integration

The system is going to need its own electrical panel. The panel should have circuit breakers, contactors, overload relays, emergency stop relays, a 24 V DC power supply, PLC, VSD, and terminal blocks. The motor supply should be kept separate from the control supply to cut down on electrical noise. The PLC and server should be hooked up to a UPS so tracking data doesn't get lost during short power cuts [9].

4.3 Network Integration

RFID readers, staff dashboard PCs, and the server will connect through Ethernet. The system should store data locally first, then sync with external systems when the internet is available. This lowers the risk of the system failing when the internet goes down.

4.4 Software Integration

The software will connect three main parts:

- RFID reader software reads the bag ID.
- Database software stores the baggage status.
- SMS software sends a message when the bag hits important checkpoints.

The staff dashboard will show colour-coded statuses like "checked in," "screened," "sorted," "loaded," "arrived," and "missing scan."

5. Design Calculations and Checks

5.1 Throughput Check

The required throughput from Assignment 1 is: 1500bags/hour

The selected carousel sorter can handle about 2700 bags/hour

Therefore, the capacity margin is:

$$2700-1500 = 1200 \text{ bags/hour}$$

Percentage spare capacity:

$$\frac{1200}{1500} \times 100 = 80\%$$

5.2 Tracking Time Check

The design says tracking must happen within 5 seconds. Since RFID portal readers scan automatically when the bag goes through the reader zone, it should scan below the limit if the antenna is positioned correctly and tested.

5.3 Communication Time Check

Passenger communication must be under 60 seconds. The target is to send SMS within 15 seconds after a valid RFID scan, as it says in the Assignment 1 data flow [1]. This gives enough room, even if the SMS gateway is a bit slow.

5.4 Design Life

The control system must run 24 hours a day cause airport never stops. Mechanical parts like belts, rollers, bearings, and pushers will wear out faster than electronic stuff. The lecture notes say design life depends on loading conditions, weather exposure, material breaking down, fatigue, wear, safety needs, and cost [4].

The proposed design life is:

Table 5 Design life

Part	Expected design life	Maintenance action
Conveyor belt	3–5 years	Inspect monthly
Bearings and rollers	2–4 years	Lubricate and replace when noisy
PLC and control panel	7–10 years	Inspect every 6 months
RFID readers	5–7 years	Clean and test monthly
Server/UPS battery	3–5 years	Replace battery when capacity drops

6. Safety Considerations

Safety matters a lot because baggage handling systems have moving conveyors, spinning motors, pushers, electrical panels, and people working around them. The lecture notes say good design should first get rid of hazards, then add protection, then warnings, then training, and lastly PPE [3].

6.1 Hazard Elimination

The design cuts down risk by putting fixed guards around moving belts, covering drive chains, and enclosing electrical panels. Walkways and maintenance areas have to stay clear. Sharp edges on frames and chutes must be removed.

6.2 Protective Systems

The system will have:

- emergency stop buttons at operator stations
- overload protection for motors
- circuit breakers and fuses in the control panel
- safety interlocks on access doors
- photoelectric sensors to detect jammed baggage
- fault alarms on the HMI dashboard

6.3 Warnings

The HMI will show warnings like:

- "Baggage jam detected."
- "RFID reader fault."
- "Motor overload."
- "Emergency stop activated."
- "Missed checkpoint scan."

A buzzer and warning light should go off during emergency stops and major faults.

6.4 Training and PPE

Operators must be trained on normal start-up, shutdown, clearing faults, lock-out procedures, and resetting emergency stops. Maintenance staff have to wear PPE like safety shoes, gloves, and eye protection when working near mechanical or electrical parts.

6.5 Factor of Safety

The lecture notes say the factor of safety covers uncertainty in materials, loading, manufacturing, and real operating conditions. Typical values are 2 to 4 for general mechanical and electrical systems [11]. For this design, the support frame and chute structure should use a minimum safety factor of 2 cause bags can get dropped, pushed, and stacked during operation.

7. Standards and Compliance

The system should follow relevant airport, electrical, data protection, and machine safety rules. IATA gives global aviation guidance and backs baggage handling improvement, while POPIA says organisations in South Africa must protect personal information [6],[7].

Table 6 System Requirements

Area	requirement	How it is applied
Baggage tracking	IATA baggage tracking guidance	RFID checkpoints track bag movement
Personal data	POPIA	Passenger phone numbers stored securely
Electrical safety	IEC/SANS electrical practices	Protection, earthing, isolation
Machine safety	Emergency stop and guarding principles	E-stops, guards, interlocks
Documentation	Engineering design documentation	Drawings, test sheets, manuals

8 Testing Plan

Got to test everything before the system gets the green light. Lecture 14 explains that testing is about catching problems, checking if performance is good enough, making the design better, keeping things safe, and getting people to trust the system [5]. Also says your test plan needs to cover what you're trying to achieve, how you're going to do it, what equipment you need, the conditions you're testing under, what counts as a pass, what could go wrong, and keeping records of everything [5].

Table 7 System Testing Plan

Test	Purpose	Acceptance criteria
Functional test	Check conveyor, sorter, pusher, RFID, and SMS	All functions work correctly
RFID scan test	Confirm bag detection	Scan success $\geq 99\%$ during test
Throughput test	Check bags per hour	≥ 1500 bags/hour
SMS test	Check notification time	SMS sent within 60 seconds
Safety test	Check emergency stops and guards	Motor stops immediately
Jam detection test	Check photocells and alerts	Jam alarm shown on HMI
Power failure test	Check UPS and recovery	Data not lost
Reliability test	Run the system for a long period	No repeated critical faults

9. Maintenance Plan

The system must be easy to maintain cause airports can't afford long downtime. Maintenance should be split into daily, monthly, and yearly tasks.

Table 8 System Maintenance Plan

Period	Maintenance work
Daily	Check conveyor movement, alarms, and visible damage
Weekly	Clean RFID readers and inspect sensors
Monthly	Check belt tension, rollers, pushers, and emergency stops
Every 6 months	Inspect the control panel, wiring, UPS, and motor protection
Yearly	Full system test, database backup, and component review

spare parts should include RFID tags, one RFID reader, photoelectric sensors, relays, fuses, roller bearings, and one spare VSD.

10. Conclusion

This report showed the detailed design of a smart airport luggage sorting and tracking system using a carousel sorter, RFID tracking, PLC control, and SMS notification. The system works well for developing airport environments cause its simpler and cheaper than a cross-belt sorter while still hitting the required throughput. The setup includes mechanical handling, RFID checkpoints, PLC control, local database storage, SMS communication, staff dashboard, and safety systems. The design also works with existing airport infrastructure by reusing conveyors where possible and adding RFID readers, control panels, and network connections. Safety is covered with guarding, emergency stops, overload protection, alarms, training, and PPE. Testing and

maintenance plans are included to make sure the system is reliable, safe, and ready to go.

References

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